REPRESENTABILITY CRITERION FOR SERRE FUNCTORS

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Allowed standard prerequisites (editorial).

- Yoneda lemma. categories-lemma-yoneda (cross-chapter, named). Representing objects determine the functor S.

Source reference. Original reference 001P: focused-seeds/geometry/S9-eqv-Serre-functor-exists/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let $\mathcal{T}$ be a k-linear triangulated category such that $\dim_k \mathrm{Hom}_{\mathcal{T}}(X, Y) < \infty$ for all $X, Y \in \mathrm{Ob}(\mathcal{T})$. The following are equivalent*

1. *there exists a k-linear equivalence $S : \mathcal{T} \to \mathcal{T}$ and k-linear isomorphisms $c_{X,Y} : \mathrm{Hom}_{\mathcal{T}}(X, Y) \to \mathrm{Hom}_{\mathcal{T}}(Y, S(X))^\vee$ functorial in $X, Y \in \mathrm{Ob}(\mathcal{T})$,*
2. *for every $X \in \mathrm{Ob}(\mathcal{T})$ the functor $Y \mapsto \mathrm{Hom}_{\mathcal{T}}(X, Y)^\vee$ is representable and the functor $Y \mapsto \mathrm{Hom}_{\mathcal{T}}(Y, X)^\vee$ is corepresentable.*

Proof. Condition (1) implies (2) since given (S, c) and $X \in \mathrm{Ob}(\mathcal{T})$ the object $S(X)$ represents the functor $Y \mapsto \mathrm{Hom}_{\mathcal{T}}(X, Y)^\vee$ and the object $S^{-1}(X)$ corepresents the functor $Y \mapsto \mathrm{Hom}_{\mathcal{T}}(Y, X)^\vee$.

Assume (2). We will repeatedly use the Yoneda lemma, see Categories, Lemma 001P. For every X denote $S(X)$ the object representing the functor $Y \mapsto \mathrm{Hom}_{\mathcal{T}}(X, Y)^\vee$. Given $\varphi : X \to X'$, we obtain a unique arrow $S(\varphi) : S(X) \to S(X')$ determined by the corresponding transformation of functors $\mathrm{Hom}_{\mathcal{T}}(X, -)^\vee \to \mathrm{Hom}_{\mathcal{T}}(X', -)^\vee$. Thus S is a functor and we obtain the isomorphisms $c_{X,Y}$ by construction. It remains to show that S is an equivalence. For every X denote $S'(X)$ the object corepresenting the functor $Y \mapsto \mathrm{Hom}_{\mathcal{T}}(Y, X)^\vee$. Arguing as above we find that S' is a functor. We claim that S' is quasi-inverse to S. To see this observe that

$$\mathrm{Hom}_{\mathcal{T}}(X, Y) = \mathrm{Hom}_{\mathcal{T}}(Y, S(X))^\vee = \mathrm{Hom}_{\mathcal{T}}(S'(S(X)), Y)$$

bifunctorially, i.e., we find $S' \circ S \cong \mathrm{id}_{\mathcal{T}}$. Similarly, we have

$$\mathrm{Hom}_{\mathcal{T}}(Y, X) = \mathrm{Hom}_{\mathcal{T}}(S'(X), Y)^\vee = \mathrm{Hom}_{\mathcal{T}}(Y, S(S'(X)))$$

and we find $S \circ S' \cong \mathrm{id}_{\mathcal{T}}$. □

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